

U.S. Airline Performance & Delay Analysis

Using Python, SQL, Power BI

1. Introduction

The aviation industry generates millions of flight records every year. Analyzing flight operations is essential for improving airline performance, reducing delays, optimizing airport operations, and enhancing passenger satisfaction.

This project was developed using the U.S. Domestic Flights dataset containing over 5.8 million flight records. The objective was to transform raw aviation data into meaningful business insights through data cleaning, SQL analysis, data modeling, and interactive dashboard visualization.

The analysis focuses on flight delays, cancellations, airline performance, airport efficiency, and operational trends.

2. Problem Statement

The objective of this project is to analyze airline operations, flight delays, cancellation patterns, airport performance, and delay causes using Python, SQL Server, and Microsoft Power BI.

The project aims to transform raw flight data into interactive dashboards and analytical reports that support operational decision-making and performance improvement within the aviation industry.

3. Project Approach

The project followed an end-to-end data analytics workflow using Python, SQL Server, and Microsoft Power BI.

Workflow:

- Cleaned and preprocessed flight datasets using Python (Pandas)
- Imported cleaned datasets into SQL Server
- Performed business analysis using SQL queries and aggregations

- Built relationships and data models in Power BI
- Developed interactive dashboards for:
 - Flight Performance Overview
 - Airline Performance Analysis
 - Airport Performance Analysis

The final dashboards were designed to generate actionable insights through KPI monitoring, trend analysis, and interactive visualizations.

4. Dataset Overview

The dataset used in this project contains U.S. domestic flight records for 2015.

The project contains three interconnected tables:

Table Name	Description	Dataset Size
Flights	Flight operational records	5,819,073 Records
Airlines	Airline reference data	14 Records
Airports	Airport reference data	322 Records

5. Data Cleaning Process

Data cleaning and preprocessing were performed using Python (Pandas).

Flights Table

Actions Performed:

- Standardized column names
- Checked and removed duplicate records
- Investigated missing values
- Replaced missing delay cause values with 0 where appropriate
- Retained operational null values for cancelled and diverted flights
- Created Flight Date column
- Created Month Name column

- Created Day Name column

Airlines Table

Actions Performed:

- Standardized column names
- Verified duplicate records
- Verified missing values
- Validated IATA codes

Airports Table

Actions Performed:

- Standardized column names
- Verified duplicate records
- Investigated latitude and longitude missing values
- Validated airport reference information

The cleaned datasets were then imported into SQL Server for business analysis and Power BI dashboard development.

6. SQL Analysis

1. What is the Overall Cancellation Rate?

```
SELECT  
ROUND(  
SUM(CAST(cancelled AS INT)) * 100.0 / COUNT(*),  
2  
) AS cancellation_rate  
FROM flights_clean;
```

Output:

	cancellation_rate
1	1.540000000000

Insight

The overall flight cancellation rate was approximately 1.54%, indicating that the majority of scheduled flights were successfully completed.

2. What is the Average Arrival and Departure Delay?

```
SELECT
```

```
ROUND(AVG(arrival_delay),2) AS avg_arrival_delay,
```

```
ROUND(AVG(departure_delay),2) AS avg_departure_delay
```

```
FROM flights_clean
```

```
WHERE cancelled = 0;
```

Output:

	avg_arrival_delay	avg_departure_delay
1	4.41	9.34

Insight

Average departure delays were significantly higher than arrival delays, suggesting operational inefficiencies primarily occur before takeoff.

3. Which Airlines Operated the Most Flights?

```
SELECT TOP 10
```

```
a.airline,
```

```
COUNT(*) AS total_flights
```

```
FROM flights_clean f
```

```
JOIN airlines_clean a
```

```
ON f.airline = a.iata_code
```

```
GROUP BY a.airline
```

```
ORDER BY total_flights DESC;
```

Output:

	airline	total_flights
1	Southwest Airlines Co.	1261855
2	Delta Air Lines Inc.	875881
3	American Airlines Inc.	725984
4	Skywest Airlines Inc.	588353
5	Atlantic Southeast Airlines	571977
6	United Air Lines Inc.	515723
7	American Eagle Airlines Inc.	294632
8	JetBlue Airways	267048
9	US Airways Inc.	198715
10	Alaska Airlines Inc.	172521

Insight

Southwest Airlines operated the highest number of flights, making it the busiest airline in the dataset.

4. Which Airlines Experienced the Highest Average Delays?

SELECT

a.airline,

ROUND(AVG(f.arrival_delay),2) AS avg_delay

FROM flights_clean f

JOIN airlines_clean a

ON f.airline = a.iata_code

WHERE f.cancelled = 0

GROUP BY a.airline

ORDER BY avg_delay DESC;

Output:

	origin_airport	avg_departure_delay
1	14222	89.11
2	ILG	29.57
3	13964	28.84
4	MVY	25.91
5	HYA	23.18
6	10154	22.86
7	10581	20.11
8	STC	18.69
9	OTH	17.78
10	10165	17.56

Insight

Certain airlines consistently recorded higher delay averages, indicating opportunities for schedule optimization and operational improvement.

5. Which Airports Handle the Highest Flight Volume?

SELECT TOP 10

origin_airport,

COUNT(*) AS total_flights

FROM flights_clean

GROUP BY origin_airport

ORDER BY total_flights DESC;

Output:

	origin_airport	total_flights
1	ATL	346836
2	ORD	285883
3	DFW	239551
4	DEN	196054
5	LAX	194673
6	SFO	148008
7	PHX	146815
8	IAH	146622
9	LAS	133181
10	MSP	112117

Insight

A small number of major airports handled a significant proportion of total U.S. flight traffic, highlighting their importance within the aviation network.

6. Top 10 Airports with Highest Average Departure Delay

SELECT TOP 10

origin_airport,

ROUND(AVG(departure_delay),2) AS avg_departure_delay

FROM flights_clean

WHERE cancelled = 0

GROUP BY origin_airport

ORDER BY avg_departure_delay DESC;

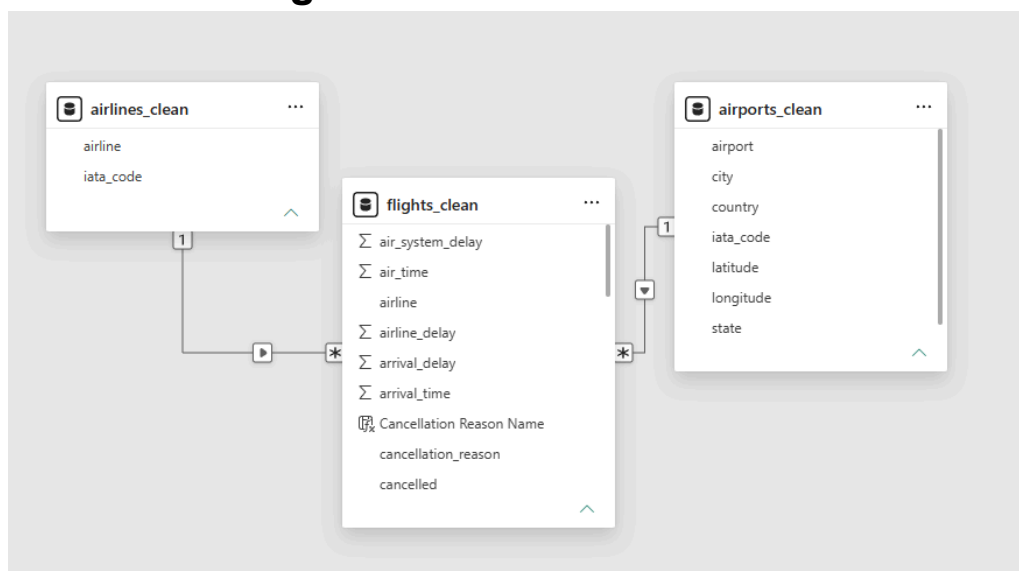
Output :

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Insights:

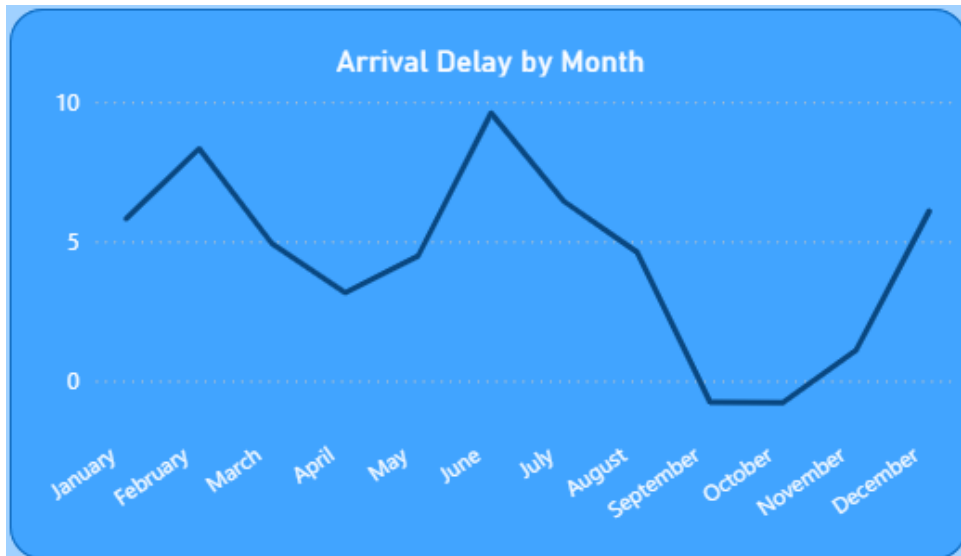
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Data Modelling



7. Important Visualizations & Insights

7.1 Arrival Delay by Month



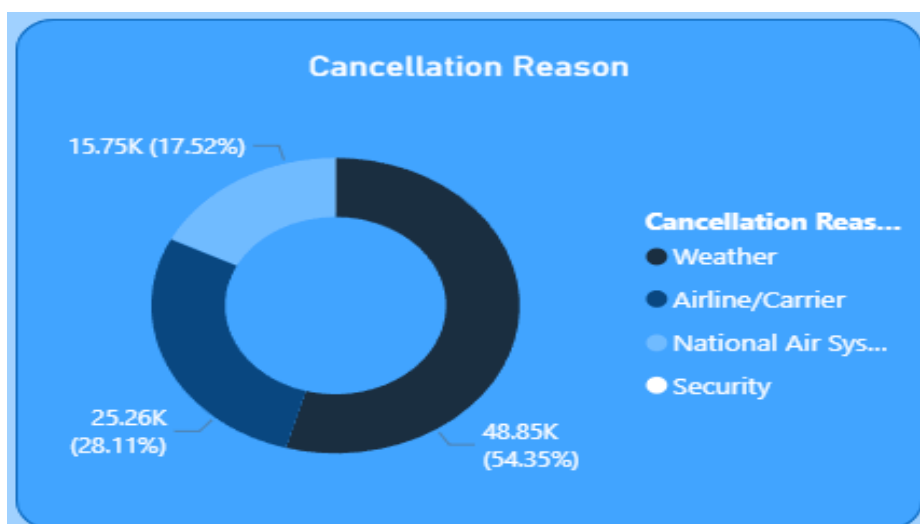
About the Visual

This line chart shows the average arrival delay across different months of the year. It helps identify seasonal patterns and periods when flight delays are more frequent.

Key Insights

- June recorded the highest average arrival delay, reaching approximately 10 minutes.
- September and October experienced the lowest average arrival delays.
- Delay levels increased again during December, indicating year-end operational challenges and increased travel demand.
- Seasonal factors significantly influence airline punctuality and operational efficiency.

7.2 Cancellation Reason Distribution



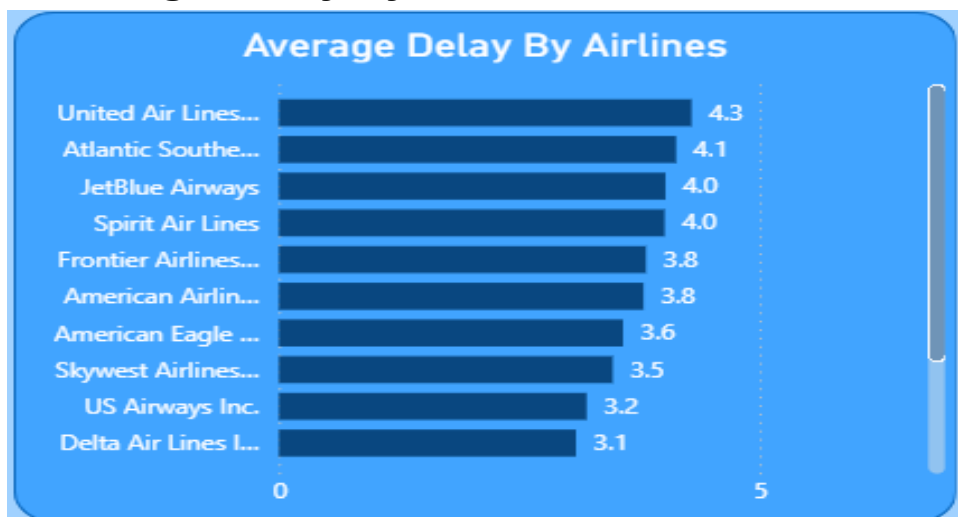
About the Visual

This donut chart displays the distribution of flight cancellations by cause, including Weather, Airline/Carrier Issues, National Air System, and Security-related disruptions.

Key Insights

- Weather-related issues accounted for the largest share of flight cancellations, contributing more than half of all cancelled flights.
 - Airline/Carrier issues represented the second-largest cancellation category.
 - Security-related cancellations were minimal compared to other operational factors.
 - External environmental conditions have a major impact on airline operations.
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7.3 Average Delay by Airlines



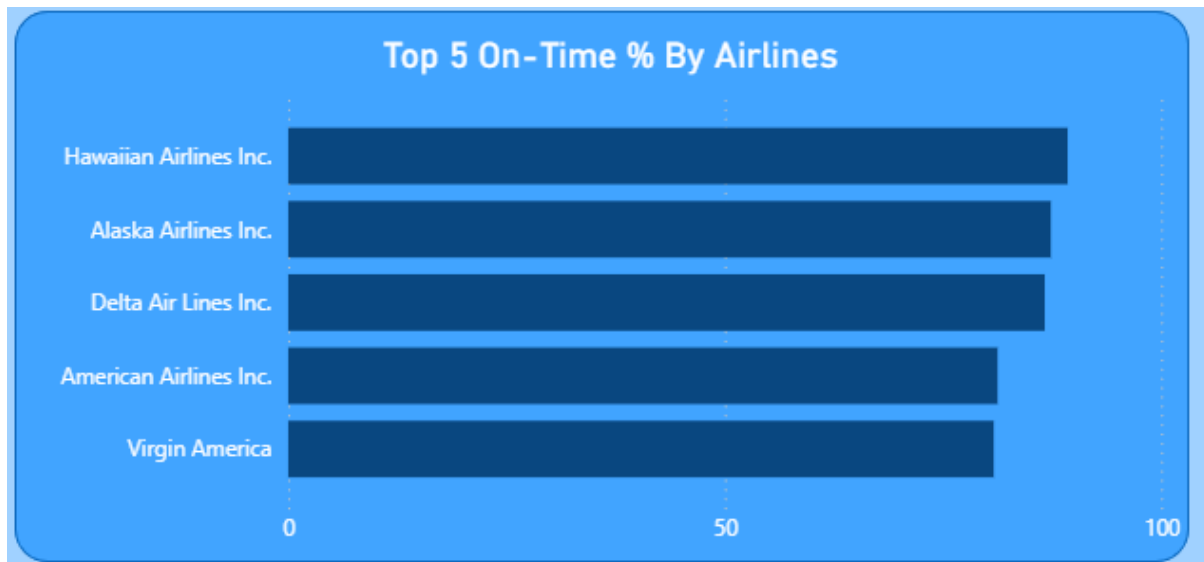
About the Visual

This bar chart compares the average arrival delay across major airlines to identify carriers experiencing higher operational delays.

Key Insights

- United Airlines recorded the highest average delay among major airlines.
 - Atlantic Southeast Airlines and JetBlue Airways also experienced relatively high delay levels.
 - Delta Air Lines demonstrated one of the lowest average delays among the airlines shown.
 - Significant performance differences exist between airlines, highlighting opportunities for operational improvement.
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7.4 Top 5 On-Time Airlines



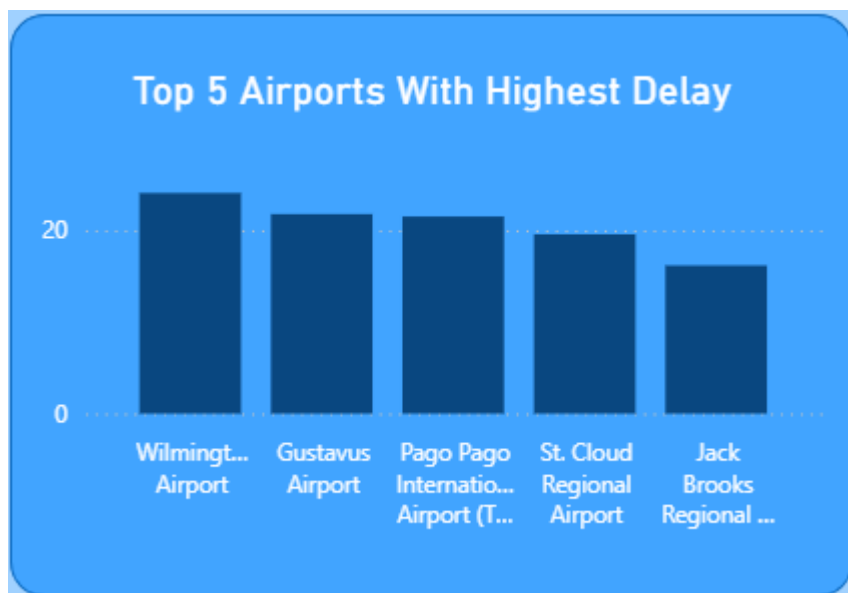
About the Visual

This chart highlights airlines with the highest on-time performance percentages, indicating the carriers that most consistently operate according to schedule.

Key Insights

- Hawaiian Airlines achieved the highest on-time performance among all airlines.
- Alaska Airlines and Delta Air Lines also maintained strong punctuality records.
- Airlines with high on-time performance demonstrate more efficient scheduling and operational execution.
- Benchmarking top-performing airlines can help improve industry-wide service quality.

7.5 Top 5 Airports with Highest Delay



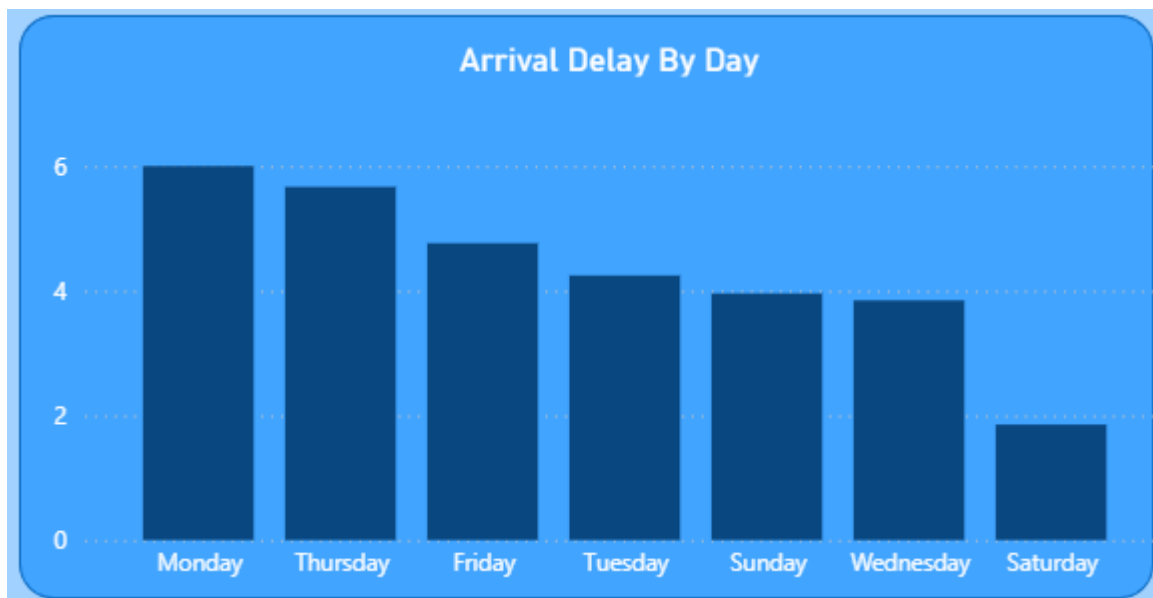
About the Visual

This visualization identifies airports with the highest average arrival delays, helping uncover locations facing operational bottlenecks.

Key Insights

- Wilmington Airport recorded the highest average delay among the analyzed airports.
 - Gustavas Airport and Pago Pago International Airport also experienced significant delays.
 - Regional airports can face unique operational challenges due to limited infrastructure and flight frequency.
 - Airport-specific operational improvements may significantly reduce overall network delays.
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7.6 Arrival Delay by Day



About the Visual

This chart compares average arrival delays across different days of the week to identify weekly delay patterns.

Key Insights

- Monday recorded the highest average arrival delay.
- Thursday and Friday also experienced relatively elevated delay levels.
- Saturday showed the lowest average arrival delay, indicating lighter operational congestion.
- Flight schedules and passenger demand patterns influence delay behavior throughout the week.

8. Business Recommendations

Based on the findings obtained from SQL analysis and Power BI dashboards, the following recommendations are proposed:

1. Improve Scheduling for High-Delay Airlines

Airlines experiencing consistently high average delays should review flight schedules, turnaround times, and crew allocation processes to minimize operational bottlenecks.

2. Strengthen Weather Preparedness Programs

Since weather-related disruptions account for the largest proportion of flight cancellations, airlines should invest in advanced weather forecasting systems and contingency planning strategies.

3. Optimize Aircraft Rotation and Maintenance Planning

Late aircraft delays contribute significantly to overall operational delays. Better aircraft utilization planning and predictive maintenance programs can reduce cascading disruptions across flight networks.

4. Improve Operational Efficiency at High-Risk Airports

Airports with consistently high average delays should focus on gate utilization, runway management, staffing optimization, and passenger flow improvements.

5. Allocate Additional Resources During Peak Delay Periods

Seasonal delay trends indicate that additional operational resources should be deployed during high-risk months such as June and December.

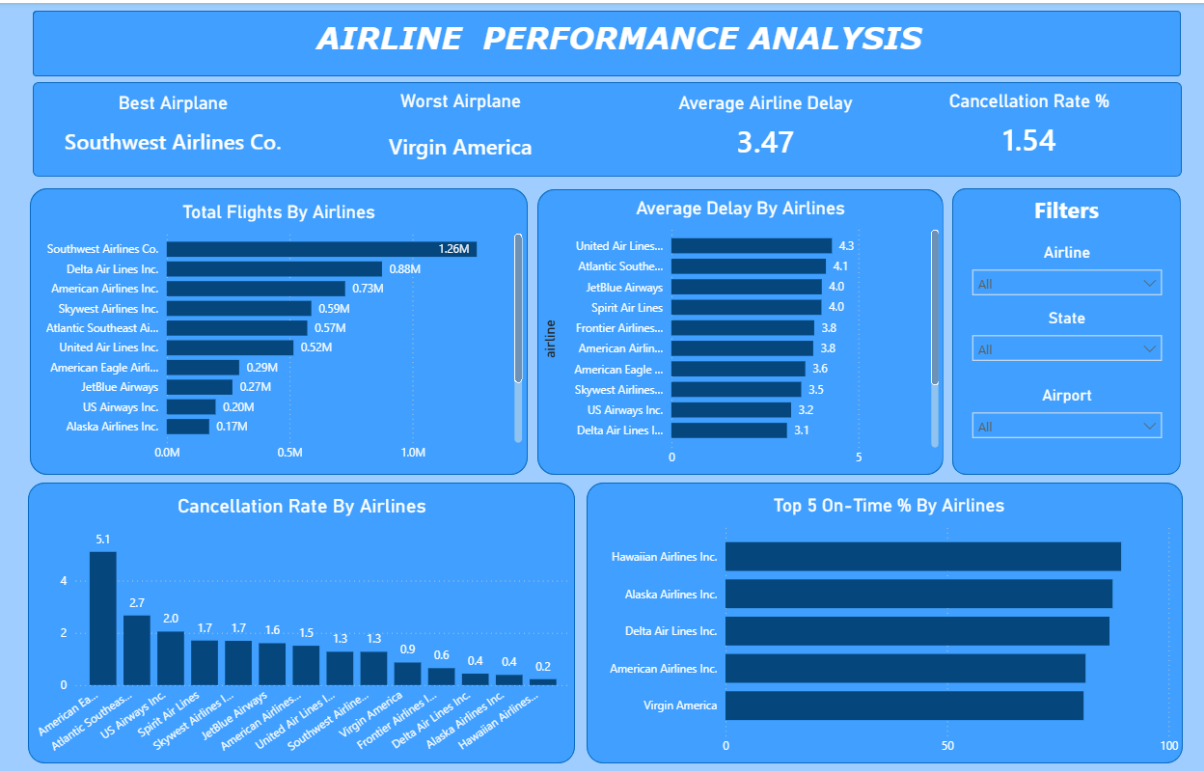
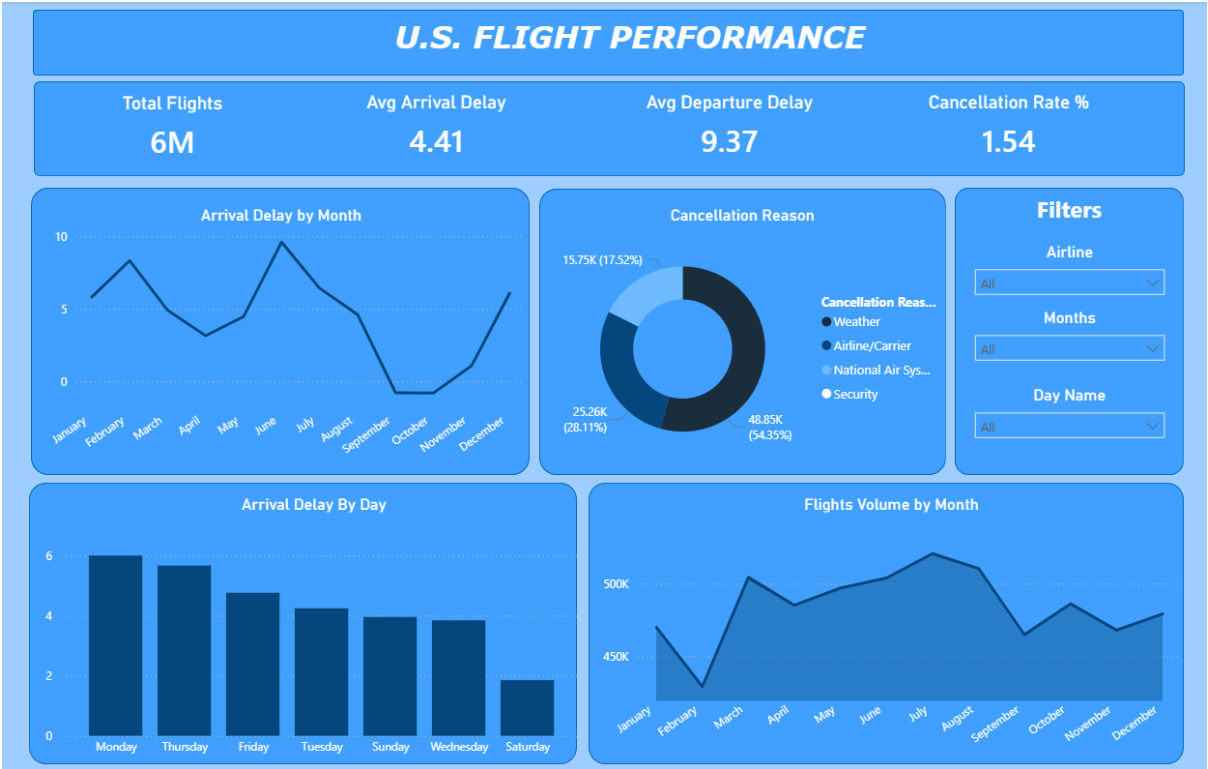
6. Benchmark Best-Performing Airlines

Airlines with strong on-time performance records can serve as benchmarks for operational best practices across the industry.

7. Implement Real-Time Delay Monitoring Systems

Real-time operational monitoring can help identify emerging delay patterns and support faster decision-making during disruptions.

9. Dashboard Images



AIRPORT PERFORMANCE ANALYSIS

Top Airport
ATL

Highest Delay Airport
Wilmington Airport

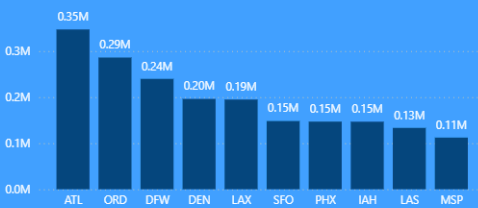
Avg Airport Delay
4.41

Avg Airport Cancellation %
1.54

Flights by Airport Location



Top 10 Busiest Airport



Filters

Airline

All

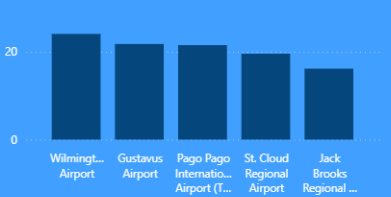
Months

All

Day Name

All

Top 5 Airports With Highest Delay



Airport Traffic vs Delay

Airports	Total Flights	Average Delay
ATL	346836	3.05
ORD	285883	8.60
DFW	239551	6.92
DEN	196054	7.19
LAX	194673	5.72
SFO	148008	5.75
PHX	146815	4.39
IAH	146622	7.38
Total	1949740	5.78